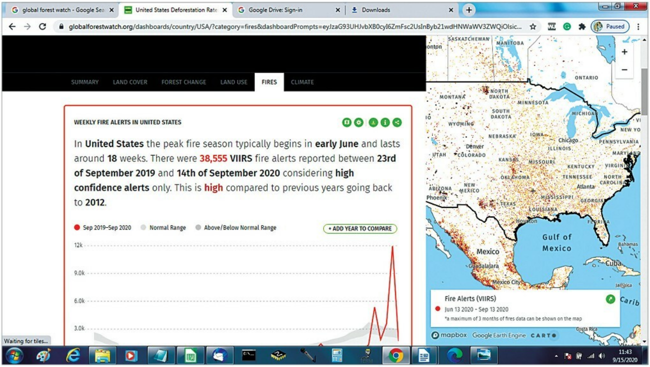

Figure 3: Forest fire alerts in the United States in 2019-20 (*Image Source: www.globalforestwatch.org*)

Figure 4: Representation of smart dustbin (*Image Source: https://earthr.org*)

Green Horizon: IBM's Green Horizon Project was first started in China. It uses AI, ML, and Big Data to create self-configuring weather and pollution forecasts. The project helps the government to enforce air quality control measures on traffic, construction, and industry. This system helps Beijing to decrease average smog levels by 35 per cent.

DeepMind: Google DeepMind developed an AI algorithm that teaches itself to use only the bare minimum amount of energy necessary to cool Google's data centres. Google was able to cut the amount of it uses to cool its data centres by 35 per cent. Studies have shown that more and more data centres are coming up, and the energy needed to power these data centres is also increasing. However, as of now, the energy usage at these centres may not be of much significance as compared to the total energy consumption.

Airlitix: Airlitix provides autonomous drone-based products for indoor agricultural and greenhouse environments. It automatically monitors crop health, manages plant inventory, and helps companies make more informed decisions. It uses AI and ML to monitor not only gases management processes but also to manage the health of national forests. It can help to collect data on temperature, humidity and carbon dioxide, as well as analyse soil and crop health.

Smart dustbins: Smart dustbins help reduce carbon emissions. Garbage trucks come for pick up only if the bins are 75 per cent full, thus saving time, energy, and reducing traffic congestion. AirBin smart dustbin (short for artificial intelligence radio bins) is one such system developed by the Indian Institute of Technology (IIT), Madras, which enables remote monitoring of waste accumulation levels through a smartphone. The system can be retrofitted on to existing garbage bins to monitor the dustbin levels.

Autonomous vehicles: Experts say that automated driving could help reduce fuel consumption by about 15 per cent. Autonomous driving provides fuel efficiency, which in turn helps reduce air pollution and global warming. END

By: Sani Theo

The author is a technical editor at EFY.

The article was originally published in the October 2020 issue of Electronics For You.